

Grade 6 Accelerated
Unit 10
Lesson 6 Homework

Name Answer Key
Date _____
Period _____

A randomly selected group of male and female students were selected to have their heights measured at their middle school. The heights, in inches, of the 15 randomly selected male and female students are shown below.

Male students: 56, 50, 49, 74, 46, 52, 61, 49, 53, 56, 60, 58, 47, 51, 59

Female students: 40, 56, 50, 60, 66, 65, 55, 54, 52, 50, 49, 62, 61, 59, 64

Complete table by using the heights of the male and female students shown above.

Measure	Male Students	Female Students
1. Mean	$\frac{821}{15} = 54.\overline{73}$	$\frac{843}{15} = 56.2$
2. Median	53	56
3. Range	$74 - 46 = 28$	$66 - 40 = 26$
4. IQR	$59 - 49 = 10$	$62 - 50 = 12$
5. Potential Outliers	74	40

6. Explain which measure you would use to describe the data sets above.

Because both male and female students have outliers, the median and IQR would give better information about the data.

7. Which of the data sets have more variability?

The IQR is larger in the heights of female students so that data has more variability.

A fast food restaurant was trying to determine which way customers could get their food the fastest, inside the lobby or in the drive thru. The restaurant paid 9 mystery shoppers to time how quickly they received their food in the lobby and in the drive thru. The times, in minutes are shown below.

	1	2	Q ₁ 2.5	3	4	Med 5	6	7	Q ₃ 7.5	8	15
Lobby	2	15		7	5	8	3	4	6		1
Drive Thru	5	3		6	5	4	6	2	7		9
	2	3	Q ₁ 3.5	4	5	Med 5	6	6	Q ₃ 6.5	7	9

8. What are the mean and median times for getting food in the lobby and in the drive thru?

	Lobby	Drive Thru
Median	5	5
Mean	$\frac{51}{9} = 5.\bar{6} = 5\frac{2}{3}$	$\frac{47}{9} = 5.\bar{2}$ or $5\frac{2}{9}$

9. Which method of getting food has the greatest IQR? Which method has the biggest range?

	Lobby	Drive Thru
IQR	$7.5 - 2.5 = 5$	$6.5 - 3.5 = 3$
Range	$15 - 1 = 14$	$9 - 2 = 7$

The lobby has the higher IQR and range.

10. Based on the times shown above, use the most appropriate measures of center and variability to determine which method you would choose to get your food from this restaurant.

The drive thru has a shorter average wait time with a smaller variability, so the drive thru is the quicker choice.